

Aim/Overview of the practical: (B). To design the circuit and write a program to interface an HC-05 Bluetooth module and a relay module with Arduino Uno, in order to wirelessly control an LED using a smartphone.

APPARATUS / COMPONENTS REQUIRED\

S.No.	Component	Specification	Qty.
1	Arduino Uno	ATmega328P board	1
2	HC-05 Bluetooth Module	Class 2, 9600 bps default	1
3	Relay Module	5V, 1-channel	1
4	LED	5 mm, any colour	1
5	Resistors	220 Ω (LED current-limiting) + 1 k Ω & 2 k Ω (voltage divider)	3
6	Connecting Wires	Male-Female / Male-Male	As required
7	Breadboard	Standard	1
8	USB Cable	Type-A to Type-B	1
9	Smartphone	With “Bluetooth Terminal” app installed	1
10	DC Power Source	5V / 9V battery or adapter (for LED circuit)	1

THEORY

Bluetooth Module

A Bluetooth module is a wireless communication device used to exchange data between a microcontroller such as Arduino Uno and another Bluetooth-enabled device, such as a smartphone, laptop, or another microcontroller. The most commonly used modules in Arduino projects are HC-05 and HC-06.

HC-05 Bluetooth Module

The HC-05 is a low-cost Bluetooth Serial Port Protocol (SPP) module used to provide wireless serial communication between a microcontroller and another Bluetooth-enabled device, such as a smartphone, tablet, or computer. It is commonly used with Arduino Uno for wireless control and monitoring applications.



Working Principle

The HC-05 converts serial data from the microcontroller into wireless Bluetooth data and converts received Bluetooth data back into serial data.

For example, in a Bluetooth-controlled LED experiment:

When a user sends a command such as 1 from a smartphone, the HC-05 receives the Bluetooth data and transfers it to the Arduino through its TXD pin. Arduino processes the command and activates the relay. Similarly, sending 0 causes Arduino to deactivate the relay.

Main Features

Feature	Description
Communication	Bluetooth wireless communication
Protocol	Bluetooth SPP
Interface	UART/Serial
Typical baud rate	9600 bps
Operating supply	Common breakout boards use 5 V supply
Logic level	HC-05 logic is approximately 3.3 V
Communication range	Typically around 10 m for common Class-2 modules
Operating mode	Master or Slave, depending on configuration
Frequency band	2.4 GHz ISM band
Configuration	AT commands
Applications	Wireless control, data transfer, robotics, IoT prototypes

HC-05 is a Class-2 Bluetooth module that implements the Serial Port Protocol (SPP), which lets it transparently convert wireless Bluetooth data into wired serial (UART) data and vice versa. It typically operates at a default baud rate of 9600 bps and has a range of about 10 metres.

HC-05 Pinout Configuration

Pin Number	Pin Name	Description
1	Enable / Key	This pin is used to toggle between Data Mode (set low) and AT command mode (set high). By default it is in Data mode
2	Vcc	Powers the module. Connect to +5V Supply voltage
3	Ground	Ground pin of module, connect to system ground.
4	TX– Transmitter	Transmits Serial Data. Everything received via Bluetooth will be given out by this pin as serial data.
5	RX – Receiver	Receive Serial Data. Every serial data given to this pin will be broadcasted via Bluetooth
6	State	The state pin is connected to on board LED, it can be used as a feedback to check if Bluetooth is working properly.
7	LED	Indicates the status of Module
		Blink once in 2 sec: Module has entered Command Mode
		Repeated Blinking: Waiting for connection in Data Mode
8	Button	Blink twice in 1 sec: Connection successful in Data Mode
		Used to control the Key/Enable pin to toggle between Data and command Mode

Relay Module

A relay module is an electronic switching device that allows a low-power controller such as an Arduino Uno to control a higher-current or electrically isolated load. In your HC-05 + Arduino + LED experiment, the relay acts as the switching interface between the Arduino and the LED/load.

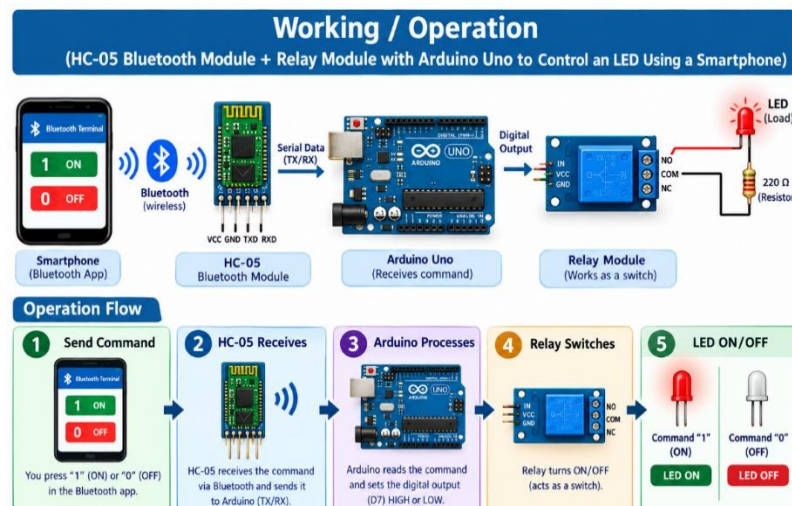
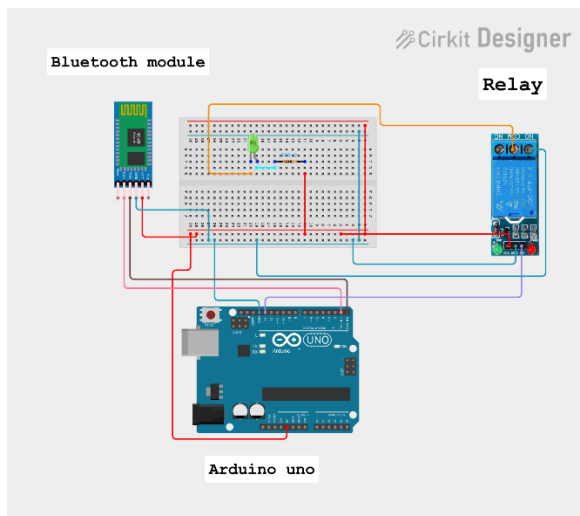
A relay is an electrically operated switch. It uses an electromagnetic coil or a solid-state switching mechanism to change the state of electrical contacts.

A conventional electromagnetic relay has:

- **Coil** – produces a magnetic field when energized.
- **COM (Common)** – common switching terminal.
- **NO (Normally Open)** – connected to COM when the relay is energized.
- **NC (Normally Closed)** – connected to COM when the relay is not energized.

A relay module is a ready-to-use circuit containing a relay and supporting components such as a transistor driver, resistor, LED indicator, and sometimes an optocoupler.

CIRCUIT DESIGN (HARDWARE INTERFACING)



Pin Connections

HC-05 → Arduino Uno

HC-05	Arduino Uno
VCC	5V/3.3V
GND	GND
TXD	D0 (RX)
RXD	D1 (TX)

Relay Module → Arduino Uno

Relay Module	Arduino Uno
VCC	5V
GND	GND
IN	D13

Relay → LED Circuit

Relay Terminal	Connection
COM	Positive terminal of DC supply (5V/9V)
NO (Normally Open)	To LED anode, through a 220Ω series resistor
LED cathode	Connected directly to DC supply negative / GND

Working Principle

The HC-05 Bluetooth module is used to establish wireless communication between the smartphone and Arduino Uno. First, the smartphone is paired with the HC-05 module. When the user sends an “ON” or “OFF” command from the Bluetooth terminal application, the HC-05 receives the command and transfers it to the Arduino Uno through UART serial communication.

The Arduino continuously checks for incoming data using the Serial.available() function. When a command is received, it is read and compared with the predefined commands “ON” and “OFF”. If the received command is “ON”, Arduino makes digital pin 13 HIGH, which activates the relay module. The relay connects its COM and NO terminals, allowing current to flow through the LED, and the LED turns ON.

When the “OFF” command is received, Arduino makes digital pin 13 LOW. The relay is deactivated, the connection between COM and NO is opened, and the LED turns OFF. Thus, the LED can be controlled wirelessly using a smartphone through the HC-05 Bluetooth module.

Working in Short

Smartphone → HC-05 → Arduino Uno → Relay Module → LED

- ON command → Relay ON → LED ON
- OFF command → Relay OFF → LED OFF

ARDUINO PROGRAM

```
#include <Arduino.h>
#include <SoftwareSerial.h>
```

```
int LED = 13;
```

```
// HC-05: Arduino RX = 10, TX = 11
```

```
SoftwareSerial Bluetooth(0, 1);
```

```
char Data;
```

```
void setup()
```

```
{  
  Serial.begin(9600);  
  Bluetooth.begin(9600);
```

```
  pinMode(LED, OUTPUT);
```

```
  // LED OFF initially  
  digitalWrite(LED, LOW);
```

```
  Serial.println("Bluetooth LED Control Ready");  
}
```

```
void loop()
```

```
{  
  if (Bluetooth.available())  
  {  
    Data = Bluetooth.read();
```

```
    // LED ON  
    if (Data == '1')  
    {  
      digitalWrite(LED, HIGH);
```

```
      Bluetooth.println("LED ON");  
      Serial.println("LED ON");  
    }
```

```
    // LED OFF  
    else if (Data == '0')  
    {  
      digitalWrite(LED, LOW);
```

```
      Bluetooth.println("LED OFF");  
      Serial.println("LED OFF");  
    }
```

```
  }  
}
```

OUTPUT



CODE EXPLANATION

1. This version does not use the SoftwareSerial library - HC-05 is wired directly to the Arduino's built-in hardware UART (Pin 0 = RX, Pin 1 = TX), so all communication uses the standard Serial object.
2. `Serial.begin(9600)` initializes the hardware serial port at 9600 bps, matching HC-05's default baud rate.
3. `pinMode(relayPin, OUTPUT)` configures Pin 13 to drive the relay, and it is initialized LOW so the LED starts in the OFF state.
4. Inside `loop()`, `Serial.available()` checks whether a new character has arrived — either from HC-05 (Bluetooth) or from the USB Serial Monitor, since both share the same hardware UART.
5. On receiving the character '1', `digitalWrite(relayPin, HIGH)` energizes the relay coil, closing the NO contact and turning the LED ON.
6. On receiving the character '0', `digitalWrite(relayPin, LOW)` de-energizes the relay, opening the contact and turning the LED OFF.
7. `Serial.println()` statements echo the current status back over the same hardware serial line for debugging purposes.

PROCEDURE

1. Connect HC-05 and the relay module to Arduino Uno as per the pin connection tables in Part A.
2. Wire the LED (with its 220Ω series resistor) to the relay's COM and NO terminals, powered from a separate 5V/9V DC supply, observing correct LED polarity.
3. Before uploading, DISCONNECT HC-05's TXD and RXD wires from Arduino Pin 0/1 (since the IDE uses the same pins through USB for programming).
4. Upload the Arduino program given above using the Arduino IDE.
5. After a successful upload, reconnect HC-05's TXD to Pin 0 and RXD to Pin 1 (through the voltage divider).
6. Power the Arduino Uno through USB or an external 5V adapter.
7. On the smartphone, enable Bluetooth and pair with HC-05 (default password usually 1234 or 0000).
8. Open a Bluetooth Terminal app on the phone and connect to the paired HC-05 device.
9. Send the character '1' from the app — the relay should click and the LED should turn ON.
10. Send the character '0' from the app — the LED should turn OFF.
11. Observe and note the corresponding status messages on the Arduino Serial Monitor.

RESULT

The HC-05 Bluetooth module and relay were successfully interfaced with Arduino Uno. The LED was controlled wirelessly (ON/OFF) from a smartphone using a Bluetooth Terminal application, confirming correct hardware connection and program execution.

PRECAUTIONS

1. Always use a voltage divider on the line feeding HC-05's RXD pin, since it is not 5V tolerant.
2. Always disconnect HC-05 from Pin 0/1 before uploading a sketch, since both the USB programmer and HC-05 share the same hardware UART — leaving it connected can cause upload failures or corrupt data.
3. Connect the LED with correct polarity and never skip the 220 Ω current-limiting resistor.
4. Double-check relay COM/NO/NC terminals before wiring the LED circuit to avoid short circuits.
5. Ensure common ground between Arduino, HC-05, and the relay module.
6. Verify the baud rate set in code (9600) matches the HC-05 module's configured baud rate.